



João Costa

Work permit: Portugal | **Date of birth:** 08/12/2001 | **Nationality:** Portuguese | **Gender:** Male | **Phone:** (+351) 935570378 (Mobile) | **Email address:** pro.joaoarcosta@gmail.com | **Website:** <https://github.com/JARCosta> | **LinkedIn:** [João Roque Costa](#) | **Whatsapp Messenger:** +351 935570378 | **Address:** Coimbra, Portugal (Home)

● ABOUT MYSELF

A Master's graduate with a strong interest in problem-solving, financial markets, artificial intelligence, and data science; eager to apply my theoretical knowledge in practical settings. Looking to join a dynamic team, develop new skills, and tackle real challenges in the IT industry.

● WORK EXPERIENCE

01/10/2024 - 31/07/2025 - LISBON, PORTUGAL

INVITED UNIVERSITY TEACHING ASSISTANT INSTITUTO SUPERIOR TÉCNICO

Taught and evaluated lab shifts on Databases and Data Science, covering SQL, database design and management using PostgreSQL and pgAdmin on the prior, and data treatment, visualization and interpretation using Python libraries such as Pandas and Matplotlib on the posterior.

01/06/2018 - 30/06/2018 - COIMBRA, PORTUGAL

SOFTWARE DEVELOPMENT SUMMER INTERN MARIA ADELAIDE AND PLAYSKEETCH

Assisted the development of an email automation system, improving workflow efficiency.

After completing the work, a game development course was awarded, where I built a Space Invaders clone and a Number Wizard game for mobile devices.

● EDUCATION & TRAINING

01/09/2023 - 31/10/2025 - LISBON, PORTUGAL

MASTERS DEGREE IN COMPUTER SCIENCE AND ENGINEERING- INSTITUTO SUPERIOR TÉCNICO

Focused on text Artificial Intelligence and Data Science, developed advanced skills in Machine Learning, Deep Learning, Autonomous Agent and Multi-Agent Systems, Data Visualisation, Data Science, and Cloud Computing, but also Optimisation and Algorithms.

To finalise the studies, I undertook a research-oriented thesis on Maximum Variance Unfolding on Disjoint Manifolds, which enlarged the method's dimensionality reduction capability into previously infeasible domains. Additionally, a publication for the International Conference on Learning Representations (ICLR) 2026 was submitted.

Some technologies and tools I worked with include: AWS, Python (Sklearn, Pandas, Numpy, Matplotlib, TensorFlow, Keras), SQL (PostgreSQL, pgAdmin, Server Management), CmapTools, Pentaho, ArchiMate, Camunda Modeler, and Enterprise Architecture (Sparx).

Field(s) of study: Software and applications development and analysis | **Level in EQF:** 7 | **Thesis:** Maximum Variance Unfolding on disjoint manifolds | **Website:**

<https://tecnico.ulisboa.pt/en/education/courses/masters-programmes/computer-science-and-engineering/>

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING- INSTITUTO SUPERIOR TÉCNICO

Worked on various projects that strengthened both my technical expertise and teamwork skills. Key areas of study and practical experience include: Mathematics and Foundations (Calculus, Physics, Algebra, Discrete Mathematics, Statistics), Low-Level Computing (Distributed Systems, Computer Networks, Operating Systems), Databases, Machine Learning, and Management.

The principal technologies and tools I utilised during my studies were: PostgreSQL, pgAdmin, Python (NumPy, Pandas), C, C++, and Java.

This experience taught me how to effectively distribute tasks, collaborate on complex projects, and develop persistence in problem-solving.

Field(s) of study:Software and applications development and analysis | **Level in EQF:** 6 | **Website:**

<https://tecnico.ulisboa.pt/en/education/courses/undergraduate-programmes/computer-science-and-engineering/>

● **SKILLS**

prepare lesson content | provide lesson materials | provide assistance to lecturer | assist students in their learning | give constructive feedback

Programming

Python (computer programming) | Java (computer programming) | C++ | C# | SQL | PostgreSQL | cloud technologies | use object-oriented programming | JavaScript

Software

SQL Server | SQL (postgresql). PgAdmin4 | Pentaho Data Integration | Archimate Modeling | Enterprise Architecture | Camunda Modeler BPMN | Git/Github, Docker, Gitlab

● **LANGUAGE SKILLS**

Mother tongue(s): **PORTUGUESE**

Other language(s): **ENGLISH**

● **DRIVING LICENCE**

Driving licence: B